

The abbreviation „AN-1” denotes the following reference:

Analysis-1 Lecture Schemes (with Homework), written by István Csörgő

It can be found in the Digital Library of the Faculty of Informatics or in the Canvas at the subject „Analysis-1”

Remember that $\mathbb{N} = \{1, 2, 3, \dots\}$.

week 1 (9-13 of February, 2026):

1. (to Lesson 2 in „AN-1”)

Determine whether the following $\mathbb{R} \rightarrow \mathbb{R}$ type functions are invertible or not. If a function is invertible, determine its inverse (domain and formula).

a) $f(x) = \frac{1}{1 + |x - 1|}$

b) $f(x) = \frac{2x + 1}{3x - 2}$

c) $f(x) = \frac{x + 1}{3x - 2} \quad D_f = [2, +\infty)$

d) $f(x) = x^2 + 4x \quad D_f = [-1, +\infty]$

Homework to this topic: „AN-1” p. 27, ex. 2

2. (to Lesson 2 in „AN-1”)

Determine the compositions $f \circ g$ and $g \circ f$ of the following $\mathbb{R} \rightarrow \mathbb{R}$ type functions if they exist (domain and formula):

a) $f(x) = \sqrt{x + 1}$
 $g(x) = x^2 - 3x + 1$

b) $f(x) = \frac{1}{x^2 - 4}$
 $g(x) = \sqrt{x - 1}$

Homework to this topic: „AN-1” p. 27, ex. 3

week 2 (16-20 of February, 2026):

3. (to Lesson 1 in „AN-1”)

Determine (without using the concept of the limit) $\sup H$, $\inf H$, $\max H$, $\min H$ if

a) $H = \left\{ \frac{n + 1}{2n + 3} \mid n \in \mathbb{N} \right\}$

b) $H = \{ \sqrt{x + 1} - \sqrt{x} \mid x \geq 0 \}$

c) $H = \left\{ \frac{1}{x} \mid 0 < x < 1 \right\}$

d) $H = \left\{ \frac{2x^2 + 1}{5x^2 + 2} \mid x \in \mathbb{R} \right\}$

Homework to this topic:

i) „AN-1” p. 16, ex. 3

ii) Determine (without using the concept of the limit) $\sup H$, $\inf H$, $\max H$, $\min H$ if

$$H = \left\{ \sqrt{x+1} - \sqrt{x} \mid x > 0 \right\}$$

4. (to Lesson 3 in „AN-1”)

Prove by definition of the limit that

$$a) \quad \lim_{n \rightarrow \infty} \frac{n}{2n-3} = \frac{1}{2}$$

$$b) \quad \lim_{n \rightarrow \infty} \frac{2n^3 + 10}{n^3 + n^2 + n + 1} = 2$$

$$c) \quad \lim_{n \rightarrow \infty} \frac{1+n^2}{2+n-2n^2} = -\frac{1}{2}$$

$$d) \quad \lim_{n \rightarrow \infty} (\sqrt{n^2+1} - n) = 0$$

In each question determine a threshold index to $\varepsilon = 0,01$.

Homework to this topic: „AN-1” p. 36, ex. 1

week 3 (23-27 of February, 2026) and week 4 (2-6 of March, 2026):

5. (to Lesson 4 in „AN-1”)

Using the results of the sections 4.1 and 4.2 in „AN-1” determine the following limits if they exist:

$$a) \quad \lim_{n \rightarrow \infty} \frac{n^3 - 3n^2 + n - 1}{1 - 2n^3 + n}$$

$$b) \quad \lim_{n \rightarrow \infty} \frac{n^4 + n^2 + n + 1}{2n^5 + n - 4}$$

$$c) \quad \lim_{n \rightarrow \infty} \frac{(2-n)^7 + (2+n)^7}{(n^2 + n + 1) \cdot (2n+1)^5}$$

$$d) \quad \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n})$$

$$e) \quad \lim_{n \rightarrow \infty} (\sqrt{n^2+n} - n)$$

$$f) \quad \lim_{n \rightarrow \infty} (\sqrt{n^2+2n+3} - \sqrt{n^2-n+1})$$

$$g) \quad \lim_{n \rightarrow \infty} \frac{\sqrt{3n^2+n} - \sqrt{2n^2+3}}{\sqrt{n^2+9} - 2}$$

$$gg) \quad \lim_{n \rightarrow \infty} \frac{\sqrt{3n^2+5} - \sqrt{3n^2+3}}{\sqrt{n^2+9} - n}$$

$$h) \quad \lim_{n \rightarrow \infty} \frac{5^{n+1} + 2^n}{3 \cdot 5^n - 5^{-n}}$$

$$i) \quad \lim_{n \rightarrow \infty} \sqrt[n]{3n^5 + 2n + 1}$$

$$j) \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n+1}{2n+3}}$$

$$k) \quad \lim_{n \rightarrow \infty} \sqrt[n]{2 \cdot 5^n + 7^n}$$

$$l) \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{6^n + 2^n \cdot n!}{n!}}$$

$$m) \quad \lim_{n \rightarrow \infty} \frac{n^2 \cdot 3^n + 2^{2n}}{4^{n+1} + 2^n}$$

$$n) \quad \lim_{n \rightarrow \infty} \sqrt{\frac{(-5)^n + 7^n}{7^{n+1} + n^7}}$$

$$o) \quad \lim_{n \rightarrow \infty} \frac{(-2)^n + n}{n! + 3^n}$$

Homework to this topic: „AN-1” p. 47, ex. 1

week 5 (9-13 of March, 2026):

6. (to Lesson 5 in „AN-1”)

Determine the limits of the following recursive sequences if they exist:

$$a) \quad a_1 = 0, \quad a_{n+1} = \sqrt{2 + a_n} \quad b) \quad a_1 = \frac{1}{4}, \quad a_{n+1} = 1 - \sqrt{1 - a_n}$$

Homework to this topic: „AN-1” p. 59, ex. 1

7. (to Lesson 5 in „AN-1”)

Determine the following limits if they exist.

$$a) \quad \lim_{n \rightarrow \infty} \left(\frac{4n+3}{4n} \right)^{5n} \quad b) \quad \lim_{n \rightarrow \infty} \left(\frac{3n+1}{3n+2} \right)^{2n+3} \quad c) \quad \lim_{n \rightarrow \infty} \left(\frac{6n-7}{6n+4} \right)^{3n+2}$$

Homework to this topic: „AN-1” p. 59, ex. 2

week 6 (16-20 of March, 2026):

8. (to Lesson 5 in „AN-1”)

Prove by definition of the limit that

$$a) \quad \lim_{n \rightarrow \infty} \frac{n^4 + 5n^3 + 11n^2 - 3n - 7}{15n^3 + 7n^2 - 21n + 5} = +\infty \quad b) \quad \lim_{n \rightarrow \infty} \frac{n^4 - 5n^3 + 8n + 3}{7 - 3n^2 - 5n^3} = -\infty$$

In question a) determine a threshold index to $P = 100$. In question b) determine a threshold index to $P = -200$.

Homework to this topic: „AN-1” p. 59, ex. 3

9. (to Lesson 5 in „AN-1”)

Determine the following limits if they exist

$$a) \quad \lim_{n \rightarrow \infty} \frac{n^3 - n^2 - 12n - 7}{13n^2 + 20n + 6} \quad b) \quad \lim_{n \rightarrow \infty} (2n - \sqrt{5n^2 + 1})$$
$$c) \quad \lim_{n \rightarrow \infty} \frac{4n^3 + 6n^2 - 11n - 8}{4 - 15n - 14n^2}$$

Homework to this topic:

- i) $\lim_{n \rightarrow \infty} (3n - \sqrt{7n^2 + 2}) = ?$
 ii) $\lim_{n \rightarrow \infty} \frac{\sqrt{3n^2 + n} - \sqrt{3n^2 - 2n + 5}}{\sqrt{n^2 + 9} - \sqrt{n^2 - 2}} = ?$

week 7 (23 - 27 of March, 2026) and week 8 (30, 31 of March and 8., 9., 10 of April, 2026):

10. (to Lesson 6 in „AN-1“)

Determine the sums of the following series.

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n}$ b) $\sum_{n=0}^{\infty} \frac{(-3)^n + 2^{n+2}}{5 \cdot 5^{n+1}}$ c) $\sum_{n=1}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^{2n}} \right)$
 d) $\sum_{n=1}^{\infty} \frac{1}{n \cdot (n+1)}$ e) $\sum_{n=1}^{\infty} \frac{1}{(2n-1) \cdot (2n+1)}$

Homework to this topic: „AN-1“ p. 69, ex. 1

11. (to Lesson 6 and 7 in „AN-1“)

Determine whether the following series are convergent or not. If a series is convergent, then determine its absolute or conditional convergence.

a) $\sum_{n=1}^{\infty} \frac{n}{2n^2 + n + 5}$ b) $\sum_{n=1}^{\infty} \frac{n}{2n^3 + n + 5}$ c) $\sum_{n=1}^{\infty} \frac{n}{2n+1}$
 d) $\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$ e) $\sum_{n=1}^{\infty} \sqrt[n]{0,1}$ f) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n \cdot (1+n^2)}}$
 g) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n \cdot (n+3)}}$ h) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$ i) $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2+n+1}$
 j) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ k) $\sum_{n=1}^{\infty} \frac{(n!)^2}{2^{n^2}}$
 l) $\sum_{n=1}^{\infty} (-1)^n \cdot (\sqrt{n+1} - \sqrt{n})$ m) $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot (n+1)}{2n^2 + 5n + 2}$
 n) $\sum_{n=1}^{\infty} \frac{2^n \cdot n!}{n^n}$ o) $\sum_{n=1}^{\infty} \frac{4^n \cdot n!}{n^n}$
 p) $\sum_{n=1}^{\infty} \frac{2^n + 3^n}{2n+1}$ q) $\sum_{n=1}^{\infty} \frac{2n+1}{(-3)^n}$

Homework to this topic: „AN-1” p. 70, ex. 2 and p. 84, ex 1

week 9 (13-17 of April, 2026):

12. (to Lesson 8 in „AN-1”)

Determine the radii of convergence and the convergence sets of the following power series. If the radius of convergence is positive, then give the domain of the corresponding analytical function.

$$a) \sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n \cdot x^n \quad b) \sum_{n=1}^{\infty} \frac{(x-1)^n}{n^p} \quad (p \in \mathbb{R} \text{ is a parameter})$$

$$c) \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} \cdot (x+2)^n \quad d) \sum_{n=0}^{\infty} \frac{4^n + (-3)^n}{n+1} \cdot x^n$$

Homework to this topic: „AN-1” p. 93, ex. 1

13. (to Lesson 8 in „AN-1”)

Expand the following functions into power series around the center $x_0 = 0$. Give the radii of convergence.

$$a) f(x) = \frac{1}{x-5} \quad b) f(x) = \frac{1}{3x-1} \quad c) f(x) = \frac{1}{2x+3}$$

$$d) f(x) = \frac{x}{x^2 - 5x + 6} \quad e) f(x) = \frac{x+3}{5x^2 + 9x - 2}$$

Homework to this topic:

i) „AN-1” p. 93, ex. 2

ii) Expand the following function into power series around the center $x_0 = 0$. Give the radius of convergence.

$$f(x) = \frac{x-20}{12x^2 + 13x - 14}$$

week 10 (20-24 of April, 2026) and week 11 (27 of April – 1 of May, 2026):

14. (to Lesson 10 in „AN-1”)

Prove by the definition of the limit that

$$a) \lim_{x \rightarrow 0} \frac{1}{1+x} = 1 \quad b) \lim_{x \rightarrow 1} \frac{x^4 + 2x^2 - 3}{x^2 - 3x + 2} = -8$$

Homework to this topic: „AN-1” p. 111, ex. 1

15. *This exercise is optional, it can be omitted (one-sided limit by definition).*
(to Lesson 11 in „AN-1”)

Prove by the definitions of the one-sided limits that

$$\lim_{x \rightarrow -3+} \frac{x^2 - 9}{|x + 3|} = -6$$

Homework to this topic: „AN-1” p. 119, ex. 1

16. (to Lessons 10-12 in „AN-1”)

Determine the following limits:

a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6}$

b) $\lim_{x \rightarrow -1} \frac{x^2 + x - 3}{x^2 + 2x + 1}$

c) $\lim_{x \rightarrow +\infty} \frac{x^3 + x - 3}{-x^2 + 2x + 1}$

d) $\lim_{x \rightarrow -\infty} \frac{x^2 + x - 3}{x^3 + 2x + 1}$

e) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$

f) $\lim_{x \rightarrow 2} \frac{x^2 + 2x - 7}{x^2 - 5x + 6}$

g) $\lim_{x \rightarrow 2} \frac{x^2 + 2x - 8}{x^2 - 5x + 6}$

h) $\lim_{x \rightarrow +\infty} \frac{2x^3 + 3x^2 + 23}{-3x^3 - 5x^2 + 31x + 1}$

Homework to this topic: „AN-1” p. 122, ex. 1

17. (to Lessons 10-12 in „AN-1”)

Determine the following limits:

a) $\lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{1 - x^2}$

b) $\lim_{x \rightarrow 0} \frac{x^2}{\sqrt[3]{1+5x} - x - 1}$

c) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 - x + 1} - \sqrt{x^2 - 1})$

Homework to this topic: „AN-1” p. 122, ex. 2

week 12 (4-8 of May, 2026):

18. (to Lesson 13 in „AN-1”)

Using the results in the examples 13.4 in „AN-1” or using power series expansion determine the following limits:

$$a) \quad \lim_{x \rightarrow 0} \frac{e^{7x} - e^{5x}}{x}$$

$$b) \quad \lim_{x \rightarrow 0} \frac{\sin(8x)}{\sin(3x)}$$

$$c) \quad \lim_{x \rightarrow 0} \frac{\sin(5x) - \sin(3x)}{\sin x}$$

$$d) \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

$$e) \quad \lim_{x \rightarrow 0} \frac{x^2}{\sqrt{1 + x \cdot \sin x} - \sqrt{\cos x}}$$

$$f) \quad \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$g) \quad \lim_{x \rightarrow \pi/3} \frac{\sin\left(x - \frac{\pi}{3}\right)}{1 - 2 \cos x}$$

$$h) \quad \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$$

$$i) \quad \lim_{x \rightarrow 0} \frac{1 - \cos^3 x + \sin^2(2x)}{2x^2 - \sin^2 x}$$

$$j) \quad \lim_{x \rightarrow 0} \frac{\cos(\sqrt{x}) - \frac{1}{1-x}}{x + \sin(2x)}$$

Homework to this topic: „AN-1” p. 126, ex. 1

week 13 (11-15 of May, 2026):

Consultations.